

Skills Battle 2026 – Planned Test Cases

Application Development | Killer Sudoku | Version 1.1

Positive	Happy path – expected normal usage
Negative	Invalid input or error conditions
Boundary	Edge values at the limits of valid input

UC-01: Read Rules

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC01-P1	Positive	None – guest visitor	Open the application URL for the first time	Welcome popup appears showing rules text, unsolved_example.png (left) and solution_example.png (right), and a 'Got it' button	UI-Manual	High
TC01-P2	Positive	rulesAcknowledged flag set in localStorage	Reload the page	Popup does NOT appear; home page shows directly	UI-Manual	Medium
TC01-N1	Negative	Server not running	Navigate to application URL	Appropriate error page shown; no crash	UI-Manual	Low
TC01-B1	Boundary	rulesAcknowledged = "" (empty string) in localStorage	Load the page	Popup appears (empty string treated as falsy)	UI-Manual	Low

UC-02: Create User (Register)

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC02-P1	Positive	No existing account	Submit valid username (6 chars), valid email, password 'Secret1!'	Account created; redirected to login page	UI-Manual	High
TC02-N1	Negative	Username 'alice' already exists in DB	Register with username 'alice'	Error: 'Username already taken'	UI-Manual	High
TC02-N2	Negative	None	Submit email 'user@' (missing domain)	Validation error: invalid email format	UI-Manual	High
TC02-N3	Negative	None	Submit password 'abc' (7 chars, no digit)	Validation error: password too short / needs digit	UI-Manual	High
TC02-N4	Negative	None	Password='Secret1!' confirmPassword='Secret2!'	Error: 'Passwords do not match'	UI-Manual	High
TC02-N5	Negative	None	Leave all fields empty and submit	All fields show required validation errors	UI-Manual	Medium
TC02-B1	Boundary	None	Username exactly 3 characters ('abc')	Accepted; account created	UI-Manual	Medium
TC02-B2	Boundary	None	Username exactly 30 characters	Accepted; account created	UI-Manual	Medium
TC02-B3	Boundary	None	Username 2 characters ('ab')	Rejected: too short	UI-Manual	Medium

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC02-B4	Boundary	None	Username 31 characters	Rejected: too long	UI-Manual	Medium
TC02-B5	Boundary	None	Password exactly 8 characters with 1 digit ('Secret1!')	Accepted	UI-Manual	Medium

UC-03: Login

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC03-P1	Positive	User 'alice' exists	Login with correct username + password	Session started; redirected to puzzle list	UI-Manual	High
TC03-N1	Negative	User 'alice' exists	Login with wrong password	Error: 'Invalid credentials' (no hint which field is wrong)	UI-Manual	High
TC03-N2	Negative	None	Login with non-existent username 'ghost'	Error: 'Invalid credentials'	UI-Manual	High
TC03-N3	Negative	None	Submit empty username and password	Validation errors on both fields	UI-Manual	Medium
TC03-B1	Boundary	User with 8-char password exists	Login with exactly that password	Accepted	UI-Manual	Medium

UC-04: Enter Puzzle

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC04-P1	Positive	User logged in	Define cages covering all 81 cells, valid sums, difficulty=2, submit	Puzzle passes to save step (UC-05)	UI-Manual	High
TC04-N1	Negative	User logged in	Leave one cell unassigned to any cage	Error: 'All cells must belong to a cage'	UI-Manual	High
TC04-N2	Negative	User logged in	Assign one cell to two different cages	Error: 'Cell already in a cage'	UI-Manual	High
TC04-N3	Negative	User logged in	Set cage target sum = 0	Validation error: sum must be >= 1	UI-Manual	High
TC04-N4	Negative	User logged in	3-cell cage with target sum = 25 (max=24)	Error: 'Sum not achievable for this cage size'	Integration	High
TC04-N5	Negative	User logged in	Set difficulty = 0	Validation error: must be 1, 2, or 3	UI-Manual	Medium
TC04-N6	Negative	User logged in	Set difficulty = 4	Validation error: must be 1, 2, or 3	UI-Manual	Medium
TC04-B1	Boundary	User logged in	Single-cell cage with target sum = 1	Accepted	Integration	Medium
TC04-B2	Boundary	User logged in	Single-cell cage with target sum = 9	Accepted	Integration	Medium
TC04-B3	Boundary	User logged in	Single-cell cage with target sum = 10	Rejected	Integration	Medium
TC04-B4	Boundary	User logged in	9-cell cage with target sum = 45 (1+2+...+9)	Accepted	Integration	Medium

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC04-B5	Boundary	User logged in	9-cell cage with target sum = 44	Rejected (no valid combo of 9 distinct digits 1-9 sums to 44)	Integration	Medium

UC-05: Save New Puzzle

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC05-P1	Positive	Valid cage layout entered (UC-04)	Submit uniquely-solvable puzzle	Puzzle saved to DB; success message shown	Unit	High
TC05-N1	Negative	Cage layout entered	Submit puzzle where cage constraints make it unsolvable	Error: 'Puzzle has no solution'; not saved	Unit	High
TC05-N2	Negative	Cage layout entered	Submit puzzle with 2+ valid solutions	Error: 'Solution is not unique'; not saved	Unit	High
TC05-B1	Boundary	Cage layout entered	Submit puzzle that is just barely solvable (one solution found on last backtrack)	Saved successfully	Unit	Medium

UC-06: Solve Puzzle

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC06-P1	Positive	Puzzle exists in DB; user logged in	Select puzzle; fill all 81 cells correctly; click Check	Solution accepted; rating prompt appears	UI-Manual	High
TC06-N1	Negative	Puzzle loaded	Type letter 'a' into a cell	Input rejected; only digits 1-9 accepted	UI-Manual	High
TC06-N2	Negative	Puzzle loaded	Type '0' into a cell	Input rejected	UI-Manual	High
TC06-N3	Negative	Puzzle loaded	Type '10' into a cell	Input rejected or truncated to first digit	UI-Manual	Medium
TC06-B1	Boundary	Puzzle loaded	Type '1' into a cell	Accepted	UI-Manual	Medium
TC06-B2	Boundary	Puzzle loaded	Type '9' into a cell	Accepted	UI-Manual	Medium

UC-07: Ask for a Hint

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC07-P1	Positive	Puzzle partially filled; user logged in	Click 'Hint' button	One correct cell value revealed; hints_used counter increments to 1	Integration	High
TC07-P2	Positive	Puzzle partially filled	Click 'Hint' 3 times	3 cells revealed; hints_used = 3	Integration	High
TC07-N1	Negative	All 81 cells already filled	Click 'Hint' button	No hint revealed; message: 'Puzzle already complete'	UI-Manual	Medium

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC07-B1	Boundary	Exactly 1 cell empty	Click 'Hint'	That last cell is revealed; puzzle now complete	Integration	Medium

UC-08: Show High Score

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC08-P1	Positive	At least one result saved in DB	Navigate to High Score page	Scores listed, sorted by score descending, showing username + time + hints + score	UI-Manual	Medium
TC08-N1	Negative	No results in DB	Navigate to High Score page	Empty state shown: 'No scores yet'; no crash	UI-Manual	Medium
TC08-B1	Boundary	Two users with identical score	View high score	Both entries shown; tie broken by solve time (earlier = higher) or shown equally	UI-Manual	Low

UC-09: Check Solution

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC09-P1	Positive	All 81 cells filled correctly	Click 'Check Solution'	'Correct!' shown; result auto-saved (UC-10)	Unit	High
TC09-N1	Negative	All 81 cells filled; one cell wrong	Click 'Check Solution'	'Incorrect' shown; wrong cells highlighted; user can continue editing	Unit	High
TC09-N2	Negative	Grid sum != 405 (quick check fail)	Click 'Check Solution'	Quick-check fails immediately; full algorithm not invoked	Unit	High
TC09-N3	Negative	Duplicate value in a row	Click 'Check Solution'	Row violation detected; error highlighted	Unit	High
TC09-N4	Negative	Cage sum does not match target	Click 'Check Solution'	Cage violation detected; cage highlighted	Unit	High
TC09-B1	Boundary	Grid sum = 405 but one cage sum wrong	Click 'Check Solution'	Quick-check passes; full algorithm catches cage error	Unit	High

UC-10: Save Result

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC10-P1	Positive	Solution verified correct	System auto-saves time=120s, hints=1	Result saved; score = max(0, 10000-120-500) = 9380	Unit	High
TC10-P2	Positive	Solution correct; hints=0, time=50s	System saves	Score = 10000-50 = 9950	Unit	High
TC10-N1	Negative	DB connection lost	Solution correct but save fails	User shown error; score displayed on screen even if not persisted	Integration	Medium

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC10-B1	Boundary	time=0, hints=0	System saves	Score = 10000 (max possible)	Unit	Medium
TC10-B2	Boundary	time=9500, hints=1	System saves	Score = max(0, 10000-9500-500) = 0	Unit	Medium
TC10-B3	Boundary	time=99999, hints=999	System saves	Score = max(0, ...) = 0 (floor at 0, never negative)	Unit	Medium

UC-11: Auto Solve

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC11-P1	Positive	Valid uniquely-solvable puzzle	Call solver	Returns correct complete 9x9 grid; isUnique=true	Unit	High
TC11-N1	Negative	Unsolvable cage constraints	Call solver	Returns null / no solution; isUnique=false	Unit	High
TC11-N2	Negative	Puzzle with 2+ valid solutions	Call solver	Returns first solution found; isUnique=false	Unit	High
TC11-B1	Boundary	80 of 81 cells pre-filled correctly	Call solver	Returns grid with the one missing cell correctly filled	Unit	High
TC11-B2	Boundary	Completely empty grid (cages only)	Call solver	Solves from scratch; returns valid solution	Unit	Medium
TC11-P2	Positive	Same puzzle solved twice sequentially	Call solver twice	Identical grid returned both times (deterministic)	Unit	Medium

UC-12: Delete Account

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC12-P1	Positive	User logged in	User clicks Delete Account; confirms in dialog	Account soft-deleted (deleted_at set); session ended; redirected to home	UI-Manual	Medium
TC12-N1	Negative	User logged in	User clicks Delete Account; cancels in dialog	Account NOT deleted; user remains logged in	UI-Manual	Medium
TC12-P2	Positive	User deleted their account	Previous results remain in DB with user_id = NULL	High score still shows anonymized entry 'Deleted User'	Integration	Low

UC-13: Puzzle Difficulty Filter

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC13-P1	Positive	Puzzles of difficulty 1, 2, 3 exist in DB	Select filter 'Easy' (difficulty=1)	Only difficulty-1 puzzles shown	UI-Manual	Medium
TC13-P2	Positive	Puzzles of all difficulties exist	Select filter 'All'	All puzzles shown	UI-Manual	Medium
TC13-N1	Negative	No difficulty-3 puzzles in DB	Select filter 'Hard'	Empty state: 'No puzzles found for this difficulty'	UI-Manual	Medium

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC13-B1	Boundary	Exactly 1 puzzle of difficulty 2	Select filter 'Medium'	That single puzzle shown	UI-Manual	Low

UC-14: Puzzle Rating

ID	Type	Precondition	Input / Action	Expected Result	Method	Priority
TC14-P1	Positive	User has completed the puzzle (result exists)	Submit rating = 4 stars	Rating saved; puzzle avg_rating updated; confirmation shown	Integration	Medium
TC14-P2	Positive	User rates puzzle, then re-rates it	Submit 3 stars, then 5 stars	Second rating replaces first (upsert); avg recalculated	Integration	Medium
TC14-N1	Negative	User has NOT solved the puzzle	Attempt to submit rating	Error: 'You must solve the puzzle before rating it'	Integration	Medium
TC14-N2	Negative	User solved the puzzle	Submit rating = 0	Validation error: must be 1-5	Unit	Medium
TC14-N3	Negative	User solved the puzzle	Submit rating = 6	Validation error: must be 1-5	Unit	Medium
TC14-B1	Boundary	User solved the puzzle	Submit rating = 1 (minimum)	Accepted; saved	Unit	Low
TC14-B2	Boundary	User solved the puzzle	Submit rating = 5 (maximum)	Accepted; saved	Unit	Low